AFM-Transformer: Adaptive CNN-Transformer for Hyperspectral Remote Sensing Image Classification

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***Abstract*—Hyperspectral image (HSI) classification plays a pivotal role in remote sensing applications. Yet, it remains a very difficult task because of the high spectral dimensionality, spatial variability, and scarcity of labeled data. To address these challenges, recent studies have explored architectures that combine convolutional neural networks (CNNs) for capturing local spatial features with Transformers for modeling long-range spectral dependencies. However, existing approaches often suffer from suboptimal feature fusion strategies that inadequately integrate the complementary information from both branches. In this study, we propose AFM-Transformer, a novel spectral-spatial dual-branch framework enhanced by an adaptive fusion module (AFM). The AFM is a lightweight, attention-based gating mechanism that dynamically learns to assign sample-specific importance to features extracted by the CNN and Transformer branches. This enables adaptive and context-aware fusion of spatial and spectral representations, enhancing the model's discriminative capability. We validate the effectiveness of the proposed AFM-Transformer through comprehensive experiments on three benchmark hyperspectral datasets. Our model consistently outperforms state-of-the-art methods in terms of classification accuracy.**

***Index Terms*—CNN, Transformer, Adaptive Feature Fusion, Hyperspectral Image Classification, Remote Sensing**

I. Introduction

With hyperspectral imaging where hundreds of contiguous spectral bands each pixel is captured, information is abundant. This spectral-spatial information has made great breakthroughs in various areas including precision agriculture, environmental surveillances, mineralogy, urban planning, and military surveillance [1]. Deciding the correct class label for every pixel is the main goal in HSI classification.

Deep learning has gained a lot of popularity today because it is capable of learning valuable and robust characteristics of data automatically, better than traditional algorithms. As a result, their outstanding performances have been evidenced in different fields of research such as HSI classification [2]. CNN-based methods have proven exceptionally effective at extracting hierarchical spatial features from HSI [3]. The fixed and localized receptive fields of CNNs however restrict its ability to detect long-range dependencies.

The Transformer architecture, originally built to support natural language processing, has recently been adapted to computer vision and HSI applications [4]. When each pixel's spectrum is treated as a sequence of tokens, Transformers can use self-attention to learn long-range relationships across the entire spectral dimension [5]. Research by Yu et al. [6] has demonstrated promising results in applying Transformer networks to HSI classification. However, compared to CNNs, Transformers lack robust inductive biases like locality and translation equivariance, leading to reduced data efficiency and higher computational costs.

Studies indicate that CNNs are very efficient in capturing local spectral-spatial characteristics, and Transformers are effective at capturing global and long-range relationships [7]. Based on the complementary strengths of these two architectures, the research community has been inspired to take hybrid CNN-Transformer models. Combining these two architectures could significantly enhance spectral–spatial feature learning. However, a critical and often overlooked challenge in this paradigm is the fusion of features from these two heterogeneous branches. Most existing methods resort to simple concatenation or element-wise addition, which are static and fail to account for the varying importance of spatial versus spectral information for different pixels. Such naive fusion can lead to suboptimal feature representations and hinder overall performance.

To mitigate these limitations, we introduce the AFM-Transformer, a dual-branch spectral–spatial model. One branch applies a CNN to find local spatial patterns, and the other branch uses a Transformer to model global spectral dependencies. At the core of this architecture is the adaptive fusion module (AFM), a lightweight, learnable gating mechanism that intelligently determines the optimal contribution

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of each branch for every input sample. Unlike static fusion methods, AFM generates adaptive weights to balance spatial and spectral information, enabling the proposed method to emphasize the most important features depending on the given context adaptively. This flexible, data-driven fusion improves the representation ability of the hybrid model and leads to better robustness in HSI classification. Major contributions of the proposed study are summarized below:

- We present the AFM-Transformer a hybrid model that integrates a CNN branch that extracts spatial features and a Transformer branch that models spectral dependencies.
- A lightweight and effective AFM module is designed, which employs a learnable gating mechanism to dynamically balance the influence of the spatial and spectral feature streams.
- We carry out extensive evaluations on three well-known HSI benchmark datasets. The AFM-Transformer offers state-of-the-art performance and has outperformed current methods.

The rest of the paper will be structured in the following manner. The related work review is presented in Section II, the proposed AFM-Transformer is presented in Section III, the results of the experiment and its analysis are presented in Section IV, and the conclusion of the study is provided in Section V.

II. Related Work

This section reviews existing methods for HSI classification, with particular attention to deep learning approaches and feature fusion strategies.

A. CNN-Based Methods

Deep learning models, particularly CNNs, have become a popular choice for HSI analysis due to their strength in learning layered feature representations and effectively modeling spatial dependencies. A more significant development came with 3D-CNNs, which employ 3D kernels to simultaneously convolve along both spectral and spatial dimensions [8]. This direct modeling of spectral-spatial cubes has shown strong performance, as demonstrated by Liu et al. [9]. Similarly, Wan et al. [10] demonstrated that leveraging both shallow and deep CNN features can improve generalization when training data is scarce. Despite their effectiveness, 3D-CNNs are parameter-intensive and rely on fixed receptive fields and struggle to capture long-range dependencies, which can limit their discriminative power in complex HSI scenarios.

B. Transformer-Based Methods

The transformer framework, originally tailored to handle linguistic data, has more recently been adapted for computer vision. The Vision Transformer (ViT) [11] encodes an image by splitting it into patch-based tokens and utilizes self-attention to learn relationships across the entire image. This idea was extended to HSI, where the spectral domain is represented as a sequential input. SpectralFormer [12] exploits self-attention to model long-range spectral dependencies, which is particularly useful for distinguishing materials with similar spectral signatures. Qing et al. [13] adapted ViT by segmenting HSI patches into smaller units, though the approach struggled to maintain local spectral-spatial correlations.

While transformers excel at capturing global context, they fall short in exploiting the structural biases of convolutional networks, such as neighborhood focus and translation invariance.

C. Feature Fusion Strategies

The effectiveness of dual-branch networks hinges on their feature fusion strategy. Primarily, fusion can occur at three different stages, such as early fusion, late fusion, and intermediate fusion. In early fusion, raw inputs or low-level features are combined at the beginning of the network. In late fusion, each branch is trained independently, and its final outputs (e.g., prediction scores) are combined. The intermediate fusion is the most common and effective strategy in modern hybrid networks. Features from different branches are combined at one or more intermediate layers. Simple fusion methods include element-wise addition or concatenation. However, these are static operations and often miss dynamic interdependencies between features.

Several recent studies have explored attention-based feature fusion, where the model learns how important each feature is during training [14]. Another powerful concept is gated fusion [15], where a learnable gate controls the flow of information from each branch. This allows for a soft, sample-dependent selection of features. Our AFM module builds on these ideas and provides a simple but effective gating strategy that adaptively balances the contributions of a CNN branch and a Transformer branch. This area has not been deeply explored in existing HSI work.

III. Proposed Methodology

In this section, the architecture of the proposed AFM- Transformer will be presented. The overall framework, depicted in Figure 1, consists of four main stages: preprocessing, a parallel dual-branch feature extractor comprising a spatial CNN branch and a spectral Transformer branch, the novel AFM module for intelligent feature integration, and a final classifier.

We denote the original hyperspectral dataset as $I \in \mathbb{R}^{H \times W \times B}$, with H and W corresponding to the spatial dimensions and B indicating the spectral dimensionality. Following standard protocols for HSI analysis, we first perform dimensionality reduction to reduce the computational complexity. Specifically, principal component analysis (PCA) is applied along the spectral dimension of I, projecting the original B-dimensional spectral vectors onto a lower-dimensional subspace spanned by the principal components that capture the highest variance. We retain the top C components, resulting in a reduced data cube $I' \in \mathbb{R}^{H \times W \times C}$, where CB, thereby effectively denoising the data while preserving the most informative spectral bands. Next, to incorporate spatial context, we extract 3D patches centered around each labeled pixel. For a pixel at position (i, j), we extract a patch $P \in \mathbb{R}^{P_s \times P_s \times C}$,

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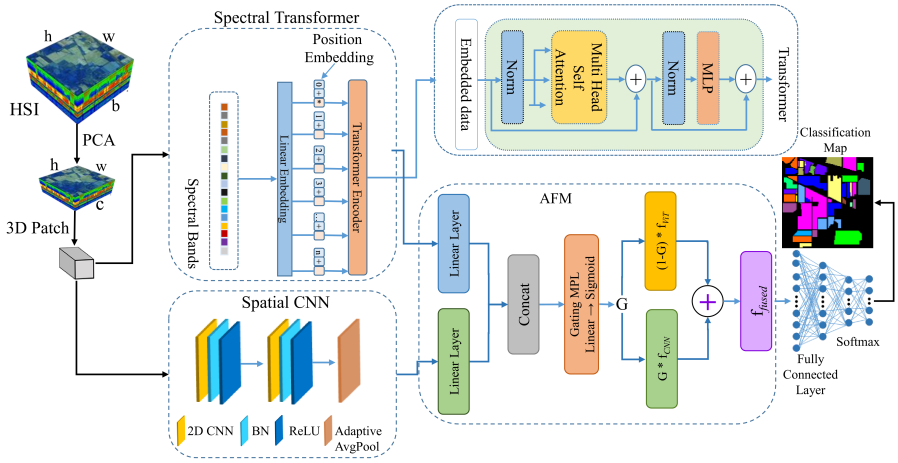

Fig. 1: Overall architecture of the AFM-Transformer.

where P_s denotes the spatial patch size. This patch is used as the input to our dual-branch network.

A. Spatial CNN Branch

The spatial branch is responsible for extracting localized spatial structures and geometric patterns from each hyperspectral patch $P \in \mathbb{R}^{P_s \times P_s \times C}$, where P_s indicates the spatial patch size. This module follows a conventional two-dimensional CNN framework, illustrated in Figure 1. Each hyperspectral patch is first interpreted as a 2D image with C channels and then passed through a sequence of convolutional stages designed to hierarchically learn spatial features. Every stage of this branch is composed of three fundamental components: a convolutional layer, batch normalization (BN), and a ReLU activation. The first block uses a 3×3 convolution with 32 filters, followed by BN and ReLU activation. While the second block increases this to 64 filters to learn more complex spatial patterns. Formally, given an input feature map $X \in \mathbb{R}^{P_s \times P_s \times C}$, the output of a single convolutional block is expressed as:

$$Y = \mathrm{ReLU}\Big(\mathrm{BN}\big(W * X + b\big)\Big), \tag{1}$$

where $W \in \mathbb{R}^{k \times k \times C_{\text{in}} \times C_{\text{out}}}$ represents the learnable convolution kernels, $b \in \mathbb{R}^{C_{\text{out}}}$ denotes the bias term, and $*$ is the convolution operator. The parameter k refers to the kernel size, which in this design is fixed at 3. The batch normalization function $\mathrm{BN}(\cdot)$ rescales each channel so that activations have zero mean and unit variance, stabilizing the training process, while $\mathrm{ReLU}(\cdot)$ introduces nonlinearity to allow the network to model more complex relationships.

Following the convolutional stages, an adaptive average pooling operation is applied, reducing the feature maps into a compact, fixed-length spatial descriptor $f_{\text{CNN}} \in \mathbb{R}^D$, with D representing the output dimensionality of the spatial branch. By stacking multiple convolutional blocks, the network learns a hierarchy of representations. This progressive learning mechanism strengthens the spatial encoding capacity of the model, thereby improving its effectiveness for hyperspectral image classification tasks.

B. Spectral Transformer Branch

The spectral branch is designed in such a way that long-range dependencies are explicitly captured, and thus, the spectral signature of the central pixel in interaction with its neighbourhood is also modeled. It is implemented using a conventional Transformer encoder architecture. For an input patch $P \in \mathbb{R}^{P_s \times P_s \times C}$, where P_s represents the spatial dimension of the patch and C denotes the number of retained principal components following PCA, we first reshape it into a sequence of P_s^2 tokens, each of dimension C, resulting in a matrix $T \in \mathbb{R}^{P_s^2 \times C}$. Each token is linearly projected into a D-dimensional embedding space to match the Transformer model dimension. To preserve spatial ordering within the patch, we add learnable positional encodings $E_{\text{pos}} \in \mathbb{R}^{P_s^2 \times D}$ to the token embeddings, producing $Z_0 = TW_e + E_{\text{pos}}$, where $W_e \in \mathbb{R}^{C \times D}$ is the projection matrix. The resulting token sequence Z_0 is fed into L standard Transformer encoder layers. The layer is made up of two main sub-modules, a multi-head self-attention (MHSA) mechanism and then a position-wise feed-forward network, and the sub-modules are connected by residual connections and layer normalization. In the MHSA module, the input sequence Z is linearly projected to produce queries (Q), keys (K), and values (V) via learned weight matrices: i.e., $Q = ZW_Q$, $K = ZW_K$, $V = ZW_V$, with $W_Q, W_K, W_V \in \mathbb{R}^{D \times d_k}$. The scaled dot-product attention weights for each head are then computed as:

$$\mathrm{Att}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \tag{2}$$

Multiple attention heads are calculated in parallel to capture diverse types of inter-band relationships, and their outputs are concatenated to form the final representation of the MHSA block. Residual skip connections and layer normalization are employed after both MHSA and feed-forward modules, ensuring training stability and efficient gradient flow. Finally, once

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the L encoder layers are processed, global average pooling is applied along the token (sequence) dimension to summarize the spectral information into a compact representation $f_{\mathrm{ViT}} \in \mathbb{R}^D$. This vector encodes long-range spectral dependencies and plays a crucial role in accurately distinguishing spectrally similar classes in HSI classification tasks.

C. Adaptive Fusion Module

In the proposed framework, the AFM is designed to dynamically integrate the spatial feature f_{CNN} and the spectral feature f_{ViT}, enabling a more effective joint representation. Unlike static fusion operations such as concatenation or addition, the AFM employs a learnable, data-dependent gating mechanism to balance the two feature streams adaptively. First, both feature vectors are projected to the same dimension D via linear layers, ensuring alignment in the embedding space. The aligned vectors are concatenated to form $[f_{\mathrm{CNN}} \,\|\, f_{\mathrm{ViT}}] \in \mathbb{R}^{2D}$, which is then passed through a lightweight gating network consisting of a fully connected layer followed by a sigmoid activation to produce the scalar gate value

$$G = \sigma\left(W_g[f_{\mathrm{CNN}} \,\|\, f_{\mathrm{ViT}}] + b_g\right), \tag{3}$$

where $W_g \in \mathbb{R}^{1\times 2D}$ and $b_g \in \mathbb{R}$ are trainable parameters, and $\sigma(\cdot)$ represents the sigmoid activation function, ensuring $G \in [0, 1]$. The final fused feature vector is obtained as

$$f_{\mathrm{fused}} = G \cdot f_{\mathrm{CNN}} + (1 - G) \cdot f_{\mathrm{ViT}}, \tag{4}$$

where G controls the contribution of the spatial features while $(1 - G)$ weights the spectral features. This adaptive mechanism enables the model to select, on a per-sample basis, the optimal balance between local spatial patterns and global spectral dependencies. For instances where spatial structures are more discriminative, G approaches 1, emphasizing f_{CNN}; conversely, when spectral signatures dominate, G tends toward 0, emphasizing f_{ViT}. This sample-wise adaptability yields a more robust and context-aware representation than is possible with static fusion strategies.

The final representation, denoted as $\mathbf{f}_{\text{fused}} \in \mathbb{R}^D$, encapsulates the adaptively aggregated spectral and spatial characteristics. The fused feature vector is subsequently fed into a lightweight classification head consisting of two fully connected (dense) layers with a final Softmax activation to obtain the probabilities of the class. The softmax function converts the network's output into a normalized probability distribution over the K land-cover categories, where the class assigned to each pixel corresponds to the category with the maximum probability value. Model training is performed by minimizing the categorical cross-entropy loss, which is the standard objective for multi-class classification problems. This loss effectively penalizes confident incorrect predictions while encouraging confident correct predictions. The cross-entropy loss is defined as:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{i,k} \log \hat{y}_{i,k} \tag{5}$$

We use N to represent all training samples in a training set in the above formulation, and K represents all the land-cover classes. The ground-truth label for the ith sample is encoded via $y_{i,k}$, where equals 1 indicates sample i belongs to class k, and 0 otherwise. Correspondingly, $\hat{y}_{i,k}$ represents the predicted probability output by the model that the ith sample belongs to class k. Minimizing this objective encourages the network to assign higher probabilities to the correct classes, thereby improving the overall discriminative power of the model.

IV. Experimental Setup and Evaluation

This section contains the detailed design and evaluation of the proposed AFM-Transformer architecture. The discussion is organized around the benchmark datasets employed, the implementation details, and the performance assessment.

A. Datasets

The model is tested on three standard HSI datasets:

- Indian Pines (IP) – 145×145 pixels, 200 bands, 16 classes
- Pavia University (PU) – 610×340 pixels, 103 bands, 9 classes
- Salinas Valley (SV) – 512×217 pixels, 204 bands, 16 classes

For PU and SV, we divided the data into 5% training, 5% validation, and 90% testing. For IP, we use 10% training, 10% validation, and 80% testing.

B. Implementation Details

For conducting experiments we used the PyTorch deep learning framework on a computational server equipped with an NVIDIA Tesla P100 GPU. The specific experimental configuration and selected hyperparameters are detailed below.

To reduce spectral redundancy while preserving the most informative features, PCA was applied as a preprocessing step. Following a combination of prior studies and our preliminary experiments, the number of retained components, C, was fixed at 30. This choice ensures that more than 99% of the spectral variability is maintained while simultaneously lowering computational demands. For the extraction of spatial context, a uniform patch size of 15×15 pixels (P_s) was employed across all datasets, which provides a reasonable balance between capturing fine-grained local structures and maintaining efficiency in training. The model parameters were optimized by the Adam algorithm, with the starting learning rate of 0.001. All the models trained through 100 epochs with a mini-batch size of 32, which was discovered to be able to settle on reliable convergence without overfitting.

We assess performance based on three common metrics, which are Overall Accuracy (OA), Average Accuracy (AA) and Kappa coefficient κ). OA measures total classification accuracy, and AA reflects the average accuracy across all classes, which provides fairness in cases where class distributions are highly unbalanced. The Kappa coefficient (κ) serves as a more robust index, quantifying the level of agreement between predictions and ground truth while adjusting for chance agreement. Together, these indicators offer a comprehensive

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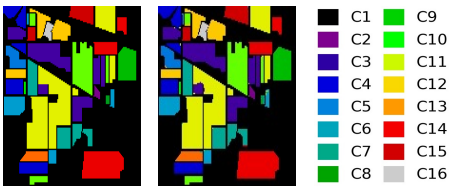
perspective on the classification capability of the proposed framework.

TABLE I: CLASSIFICATION PERFORMANCE OF PROPOSED AFM-TRANSFORMER ON THREE BENCHMARK DATASETS

Class	Indian Pines	Pavia University	Salinas Valley
C1	1.00	0.99	1.0000
C2	0.97	0.99	1.0000
C3	0.99	1.00	1.0000
C4	0.96	0.99	0.9952
C5	0.93	1.00	0.9992
C6	0.98	0.99	0.9983
C7	0.95	1.00	0.9988
C8	1.00	0.99	0.9995
C9	1.00	1.00	1.0000
C10	0.97		0.9976
C11	0.99		1.0000
C12	0.98		1.0000
C13	0.92		0.9830
C14	1.00		0.9428
C15	1.00		0.9992
C16	1.00		0.9963
OA	**98.60**	**99.91**	**99.77**
AA	**98.09**	**99.92**	**99.43**
k	**98.41**	**99.89**	**99.74**

C. Results

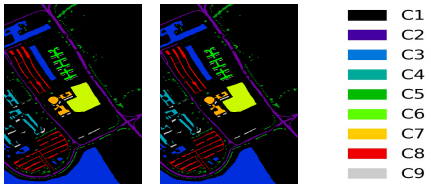
Table I summarizes the classification outcomes of the proposed AFM-Transformer on the IP, PU, and SV datasets, measured using class-wise accuracy, OA, AA, and κ. The results indicate that our approach achieves consistently strong performance across all benchmarks and evaluation measures. Specifically, on the challenging IP dataset, the AFM-Transformer attains an OA of 98.60%, an AA of 98.09%, and a κ of 98.41%, highlighting both its robustness in handling underrepresented classes and its ability to generate reliable classification maps. In the case of the PU dataset, the OA of the model is 99.91% and, on the SV dataset, 99.77%. These findings validate that the AFM effectively integrates spectral and spatial cues, leading to the extraction of highly discriminative features and resulting in state-of-the-art classification accuracy.



(a) (b) (c)

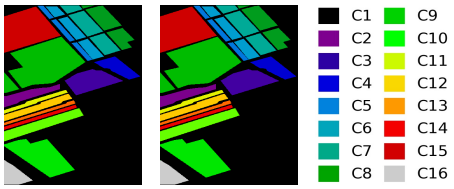
Fig. 2: IP dataset classification results. (a) Ground truth distribution of land-cover classes. (b) Predicted classification map generated by the proposed AFM-Transformer. (c) Key.

We conducted a comparative study of the proposed AFM-Transformer against several recent state-of-the-art techniques, namely SST-FA [16], Multiscale RNN [17], MV-DNNet [18], GroupFormer [19], and LSDnet [20]. Table II summarizes the overall classification of the IP, PU and SV datasets and the performance is measured by OA, AA and k. To make it clear, the highest scores of each measure are highlighted in bold. Based on Table II, it can be observed that AFM-Transformer is always able to perform better with all datasets and metrics, which provides it with an edge over the current methods.



(a) (b) (c)

Fig. 3: PU dataset classification results. (a) Ground truth distribution of land-cover classes. (b) Predicted classification map generated by the proposed AFM-Transformer. (c) Key.



(a) (b) (c)

Fig. 4: SV dataset classification results. (a) Ground truth distribution of land-cover classes. (b) Predicted classification map generated by the proposed AFM-Transformer. (c) Key.

To provide a visual assessment of the classification performance, classification maps generated by AFM-Transformer for the three datasets are displayed in Figures 2, 3, and 4. The classification maps exhibit significantly less salt-and-pepper. The boundaries between different classes are more clearly defined and align better with the ground truth. This visual evidence validates the quantitative results, indicating that the adaptively fused features are more robust to noise and spatial variability, leading to a more coherent and accurate classification.

Table III provides an evaluation of the computational efficiency of the AFM-Transformer, detailing its runtime for both training and testing.

V. Conclusion

In this study, we introduced the AFM-Transformer, an innovative spectral-spatial framework that combines a spatial CNN branch with a spectral Transformer branch via an advanced adaptive fusion module (AFM).The AFM employs a trainable, attention-based gating process to dynamically control the impact of local spatial patterns and global spectral interactions. Extensive experiments on three benchmark datasets demonstrated that the AFM-Transformer achieved an outstanding performance, being better than the existing methods in all standard evaluation measures. Qualitative results further validated the model's superiority, displaying clearer and precise classification maps. For future work, we plan to investigate

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TABLE II: CLASSIFICATION PERFORMANCE OF THE COMPARED METHODS ON THREE BENCHMARK DATASETS

Model	Indian Pines			Pavia University			Salinas Valley		
	OA	AA	k	OA	AA	k	OA	AA	k
SST-FA [16]	88.98	68.15	86.70	93.37	85.01	91.65	94.94	93.05	94.32
Multiscale RNN [17]	95.38	95.45	95.00	99.18	98.81	98.00	97.92	97.67	97.00
MV-DNNet [18]	97.68	97.55	97.62	99.16	99.04	98.07	98.24	98.17	98.00
GroupFormer [19]	97.60	94.65	97.20	98.47	97.23	97.90	99.74	**99.76**	99.70
LSDnet [20]	92.41	93.30	91.50	93.75	94.28	92.49	91.08	93.62	90.05
AFM-Transformer	**98.60**	**98.09**	**98.41**	**99.91**	**99.92**	**99.89**	**99.77**	99.43	**99.74**

TABLE III: TRAINING AND TEST TIME OF THE PROPOSED AFM-TRANSFORMER.

Dataset	Train (s)	Test (s)
Indian Pines	42.38	0.98
Pavia University	82.30	4.36
Salinas Valley	114.97	5.86

model compression strategies, such as pruning or knowledge distillation, to develop a more efficient, lightweight AFM-Transformer.

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